








# MiniPoly: Automatic Extraction of Efficient JavaScript Polyfill from Language Specification

Jihyeok Park




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# Programming Language Research Group

PLRG

- Research Field - **Programming Languages** and **Software Engineering**
  - Program Analysis
  - Mechanized Language Specification
  - Automated Testing
  - Program Synthesis & Repair
- **Recent Publications** (2020~)
  - **PLDI** (2023, 2024) / **SAS** (2026)
  - **ICSE** (2021) /
    **FSE** (2021, 2022, 2025-Demo) /
    **ASE** (2020, 2021, 2025, 2026)
  - **CACM** (2024) / **CSUR** (2021) /
    **TOSEM** (2026)



# Built-in APIs in JavaScript

**ES2025**

**Native Code**
(e.g., C++, Java)

```
const a = new Set([1, 2, 3]);
const b = new Set([1, 2]);
a.isSupersetOf(b);   // true
```



≤ 126

✗



≤ 2025.09

✗



≥ 122

✓



≤ 16.7

✗



≤ 21

✗

# Polyfill

- **Polyfill** is a **JavaScript** implementation of built-in APIs
- **core-js** is used by **41% of websites** and **≥ 46M downloads per week**

```
module.exports = function isSupersetOf(other) {
  var O = aSet(this);
  var otherRec = getSetRecord(other);
  if (size(O) < otherRec.size) return false;
  var iterator = otherRec.getIterator();
  return iterateSimple(iterator, function (e) {
    if (!has(O, e)) {
      return iteratorClose(iterator, 'normal', false);
    }
  }) !== false;
};
```

core-js

# Polyfill

```js
import 'core-js/actual/set/is-superset-of';
const a = new Set([1, 2, 3]);
const b = new Set([1, 2]);
a.isSupersetOf(b);   // true
```

```js
module.exports = function isSupersetOf(other) {
  var O = aSet(this);
  var otherRec = getSetRecord(other);
  if (size(O) < otherRec.size) return false;
  var iterator = otherRec.getIterator();
  return iterateSimple(iterator, function (e) {
    if (!has(O, e)) {
      return iteratorClose(iterator, 'normal', false);
    }
  }) !== false;
};
```

core-js



≤ 126



≤ 2025.09



≥ 122



≤ 16.7



≤ 21











# ECMA-262 - JavaScript Language Specification

**Built-in**

**Set.prototype.isSupersetOf** ( *other* )
1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < otherRec.[[Size]], return **false**.
5. Let *keysIter* be ? GetIteratorFromMethod( *otherRec*.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
   a. Set next to ? IteratorStepValue(*keysIter*).
   b. If *next* is not DONE, then
      i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
         1. Perform ? IteratorClose(*keysIter*, NormalCompletion(UNUSED)).
         2. Return **false**.
8. Return **true**.

tc39/ecma262

Status, process, and documents for ECMA-262

TC 39

**Syntax**

*AdditiveExpression*[Yield, Await] :
*MultiplicativeExpression*[?Yield, ?Await]
*AdditiveExpression*[?Yield, ?Await] + *MultiplicativeExpression*[?Yield, ?Await]
*AdditiveExpression*[?Yield, ?Await] - *MultiplicativeExpression*[?Yield, ?Await]

*AdditiveExpression* : *AdditiveExpression* + *MultiplicativeExpression*

1. Return ? EvaluateStringOrNumericBinaryExpression( *AdditiveExpression*, +, *MultiplicativeExpression*).

**Semantics**

# Current Approach - **Manual Translation**

**Built-in**

**Set.prototype.isSupersetOf** ( *other* )

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < otherRec.[[Size]], return **false**.
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      i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
         1. Perform ? IteratorClose(*keysIter*, NormalCompletion(UNUSED)).
         2. Return **false**.
8. Return **true**.

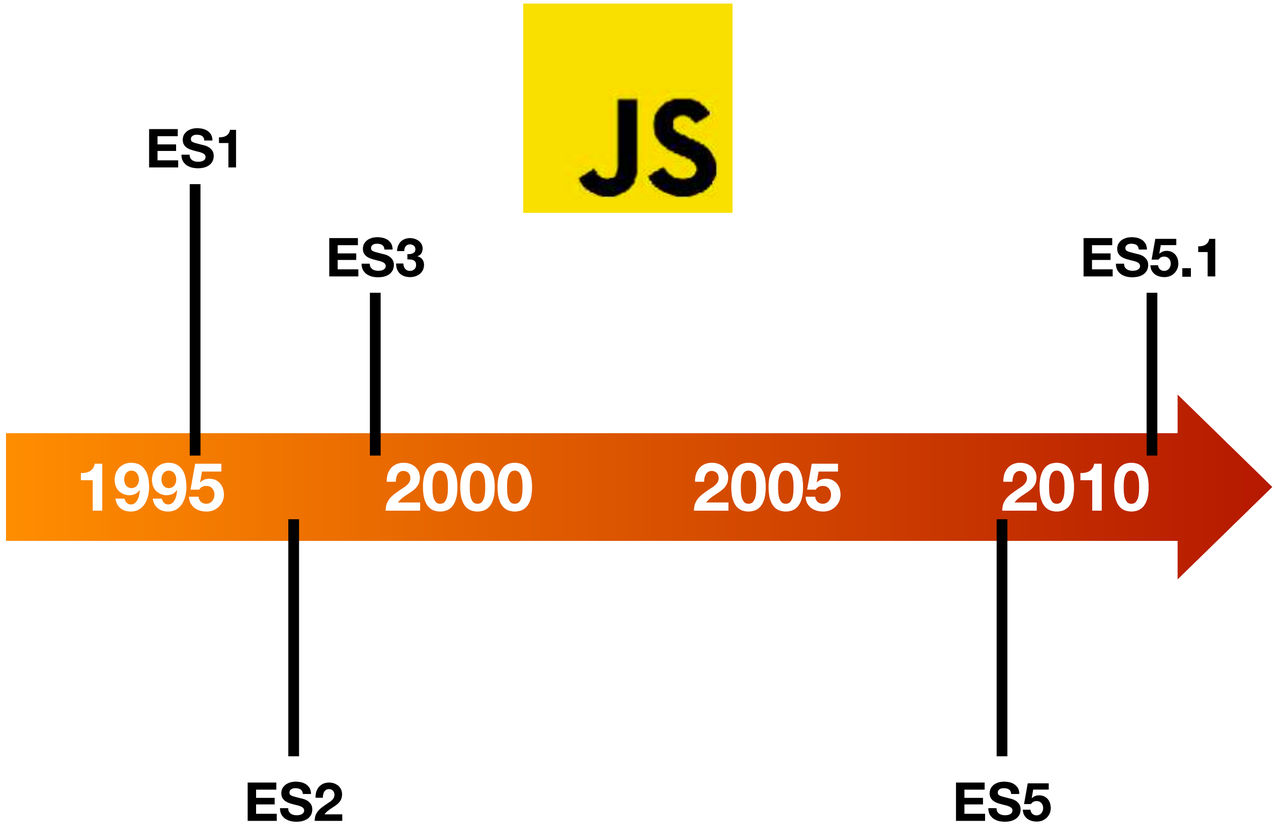
TC 39

Manual

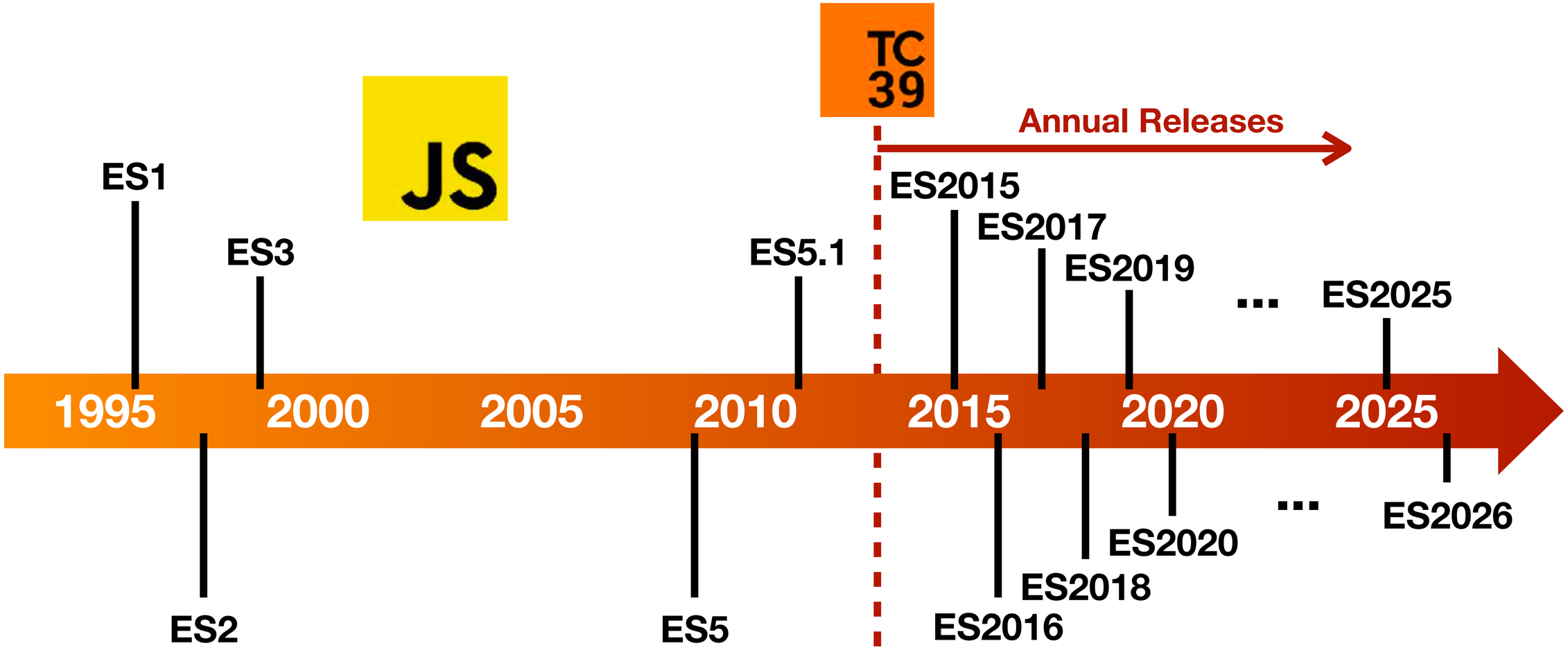
```
function isSupersetOf(other) {
  var O = aSet(this);
  var otherRec = getSetRecord(other);
  if (size(O) < otherRec.size) return false;
  var iterator = otherRec.getIterator();
  return iterateSimple(iterator,
    function (e) {
      if (!has(O, e)) {
        return iteratorClose(iterator,
                'normal', false);
      }
    }) !== false;
};
```

core-js

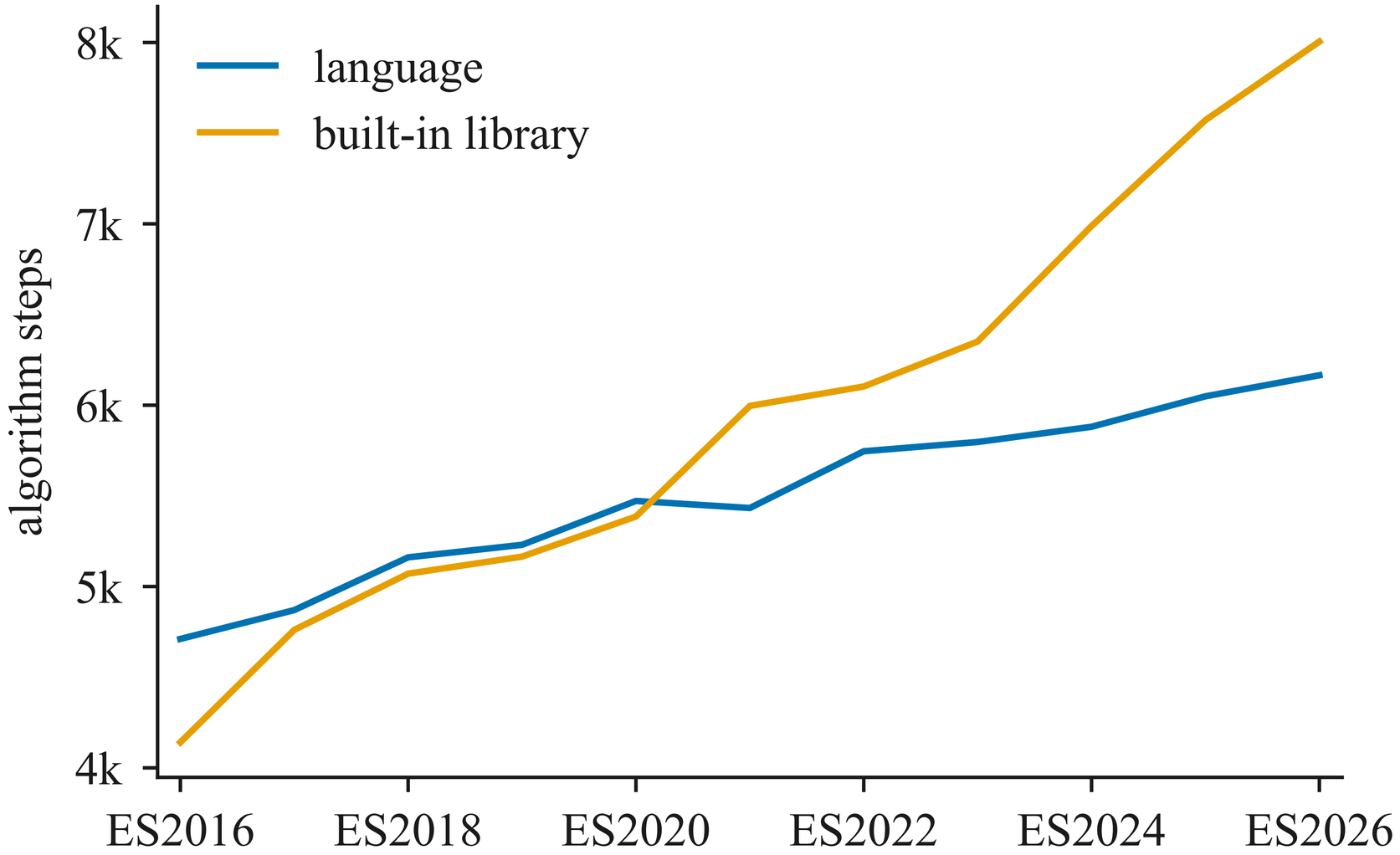
# Problem - **Fast Evolving** JavaScript



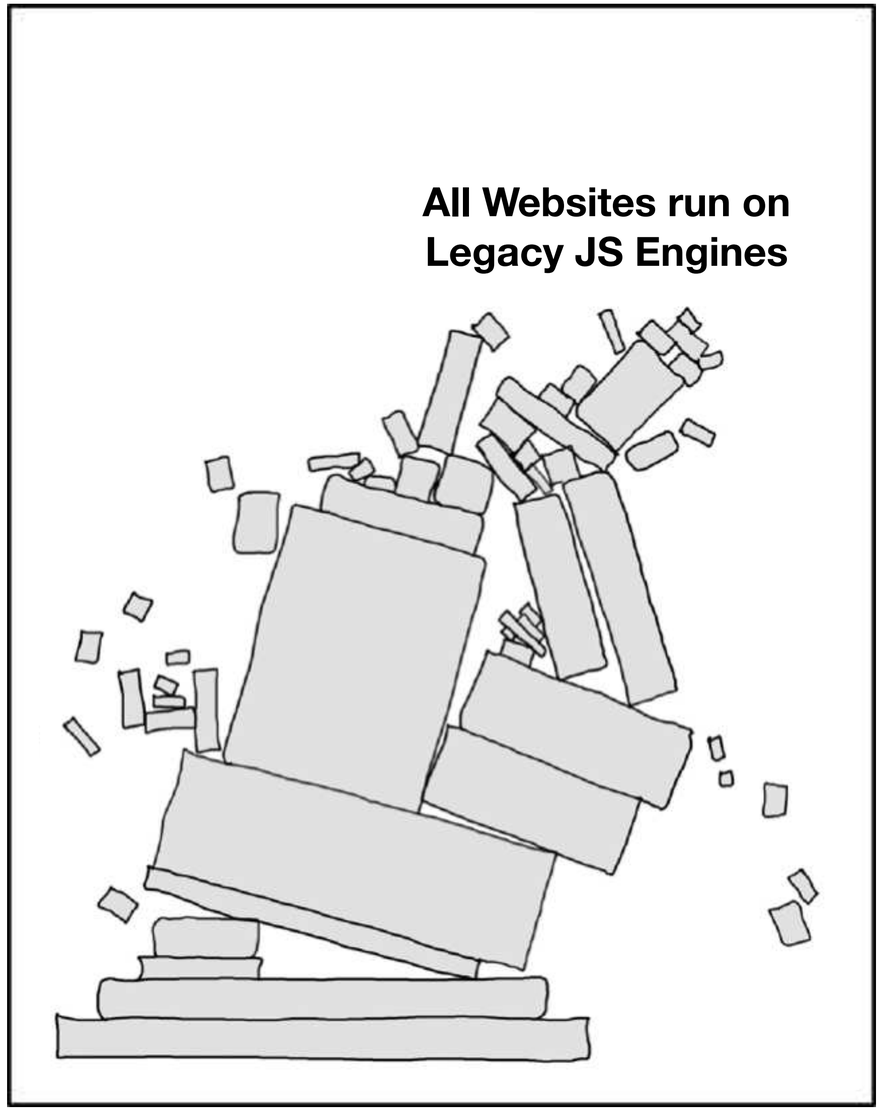
# Problem - **Fast Evolving** JavaScript

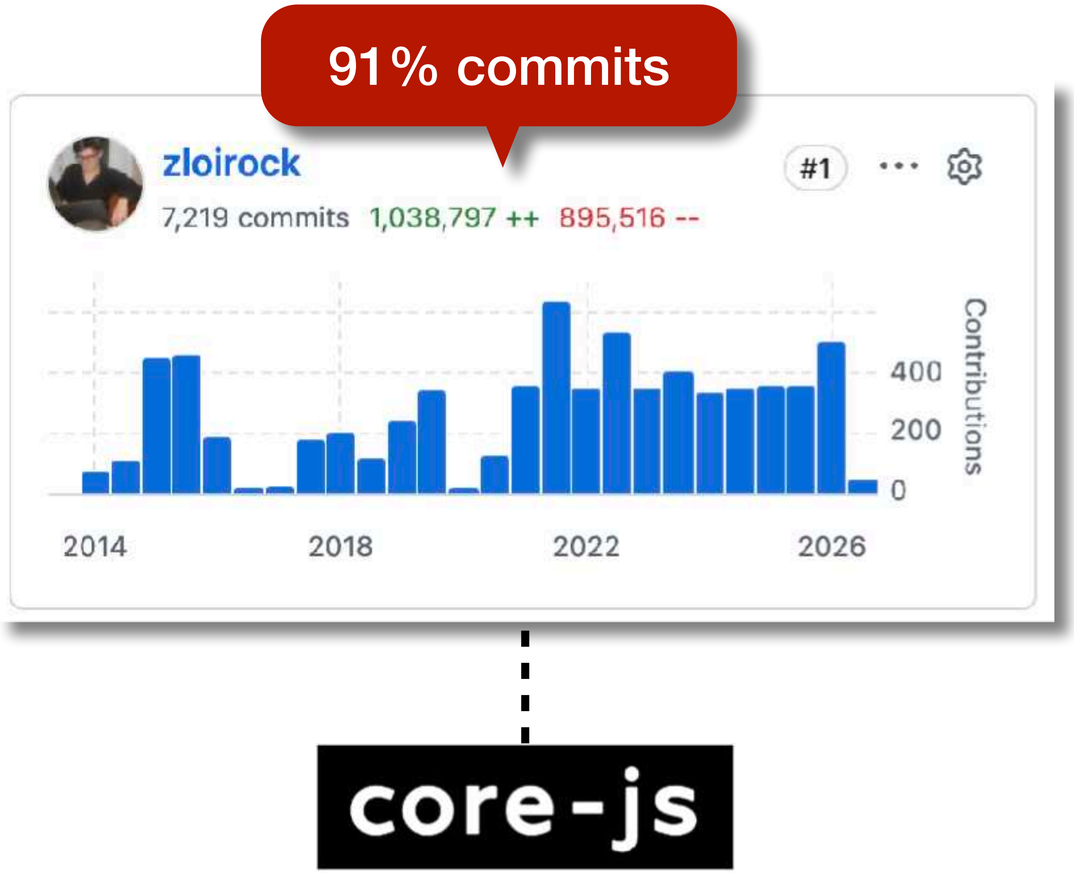


# Problem - **Fast Evolving** JavaScript



# Problem - Burden of **Manual** Implementation



# Our Approach - **Automatic Extraction** from ECMA-262

**Built-in**

**Set.prototype.isSupersetOf** ( *other* )
1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < otherRec.[[Size]], return **false**.
5. Let *keysIter* be ? GetIteratorFromMethod( *otherRec*.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
   a. Set next to ? IteratorStepValue(*keysIter*).
   b. If *next* is not DONE, then
      i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
         1. Perform ? IteratorClose(*keysIter*, NormalCompletion(UNUSED)).
         2. Return **false**.
8. Return **true**.

Extract

TC 39

**Polyfill** in **JavaScript**

# Challenges - **Correctness** and **Efficiency**

**Built-in**

**Set.prototype.isSupersetOf** ( *other* )
1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < otherRec.[[Size]], return **false**.
5. Let *keysIter* be ? GetIteratorFromMethod( *otherRec*.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
   a. Set next to ? IteratorStepValue(*keysIter*).
   b. If *next* is not DONE, then
      i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
         1. Perform ? IteratorClose(*keysIter*, NormalCompletion(UNUSED)).
         2. Return **false**.
8. Return **true**.

1) How to **correctly** handle *internal repr. of control flows*?

2) How to extract *algorithmically* **efficient** polyfill?

Extract

**Polyfill** in **JavaScript**

TC 39

# Challenge 1 - **Correctness** (Internal Repr. of Control Flow)

A computation **X**

- is **normally** terminated with **a value** — **if X** is a **normal completion**
- **throws** an **exception** — **if X** is a **throw completion**

**IteratorClose** ( *iteratorRecord*, *completion* )

...

3. Let *innerResult* be **GetMethod**(...).
4. If *innerResult* is a **normal completion**, then
   a. Let *return* be *innerResult*.[[Value]].
   b. If *return* is undefined, return ? *completion*.
   c. Set *innerResult* to **Call**(...).
5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.

...

8. Return ? *completion*.

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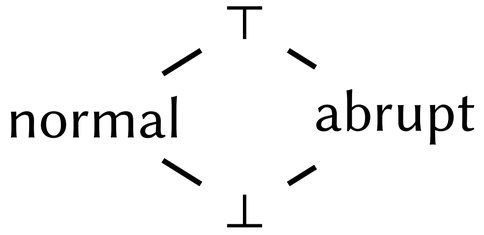
`return completion`
**OR**
`throw completion`

`throw completion`

`throw innerResult`

`return completion`

# Solution 1 - **Completion Type Analysis**




**IteratorClose** ( *iteratorRecord*, *completion* )
...
3. Let *innerResult* be **GetMethod**(...).

*completion* : ⊤
*innerResult* : ⊤

Completion

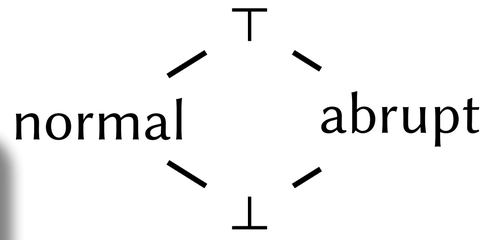
Completion

TC 39

```
function IteratorClose(iteratorRecord, completion, completion_flag) {
  ...
  try {
    var innerResult = GetMethod(...);
    var innerResult_flag = true;
  } catch(e) {
    var innerResult = e;
    var innerResult_flag = false;
  }
```

JS

# Solution 1 - **Completion Type Analysis**




4. If *innerResult* is a **normal completion**, then
   a. Let *return* be *innerResult*.[[Value]].
   b. If *return* is undefined, return ? *completion*.
   c. Set *innerResult* to **Call**(...).

*completion* : ⊤
*innerResult* : normal

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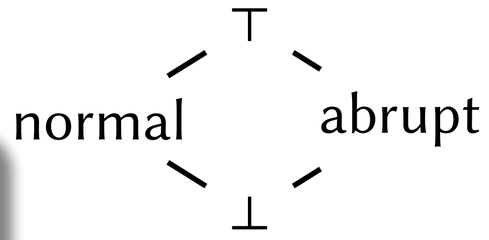
```
if (innerResult_flag) {
  ...



}
```

JS

# Solution 1 - **Completion Type Analysis**




4. If *innerResult* is a **normal completion**, then
   a. Let *return* be *innerResult*.[[Value]].
   b. If *return* is undefined, return ? *completion*.
   c. Set *innerResult* to **Call**(...).

*completion* : ⊤
*innerResult* : normal

TC 39

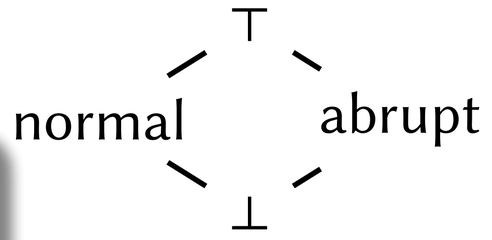
Completion

```
if (innerResult_flag) {
  ...
  if ( return === undefined) {
    if (completion_flag) return completion;
    else                 throw  completion;
  }

}
```

JS

# Solution 1 - **Completion Type Analysis**




4. If *innerResult* is a **normal completion**, then
   a. Let *return* be *innerResult*.[[Value]].
   b. If *return* is undefined, return ? *completion*.
   c. Set *innerResult* to **Call**(...).

*completion* : ⊤
*innerResult* : normal

*innerResult* : ⊤

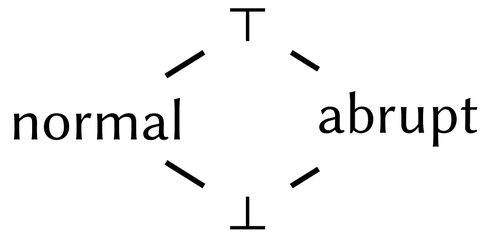
Completion

TC 39

```
if (innerResult_flag) {
  ...
  if ( return === undefined) {
    if (completion_flag) return completion;
    else                 throw  completion;
  }
  try { innerResult = Call(...); innerResult_flag = true; } catch ...
}
```

JS

# Solution 1 - **Completion Type Analysis**




5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
...
8. Return ? *completion*.

*completion* : normal

TC 39

*completion* : abrupt

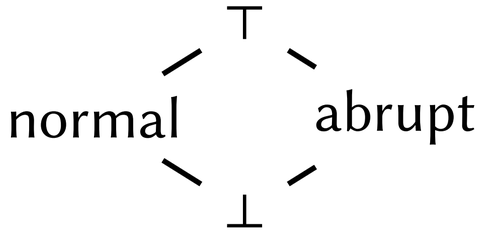
*innerResult* : abrupt

```
if (!completion_flag)  throw completion;

if (!innerResult_flag) throw innerResult;
```

JS

# Solution 1 - **Completion Type Analysis**




5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
...
8. Return ? *completion*.

*completion* : normal

TC 39

*completion* : abrupt

*innerResult* : abrupt

```
if (!completion_flag)  throw completion;

if (!innerResult_flag) throw innerResult;

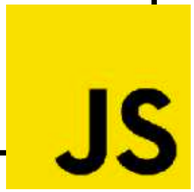
...

return completion;
```

JS

# Solution 1 - **Completion Type Analysis**

```
function IteratorClose(iteratorRecord, completion, completion_flag) {
  ...
  try { var innerResult = GetMethod(...); var innerResult_flag = true; }
  catch(e) { var innerResult = e; var innerResult_flag = false; }
  if (innerResult_flag) {
    ...
    if (_return === undefined) {
      if (completion_flag) return completion;
      else                 throw  completion;
    }
    try { innerResult = Call(...); innerResult_flag = true; } catch ...
  }
  if (!completion_flag)  throw completion;
  if (!innerResult_flag) throw innerResult;
  ...
  return completion;
}
```



# Challenge 2- **Efficiency**

$\mathcal{O}(n)$ **vs.** $\mathcal{O}(1)$

**Set.prototype.add** ( *value* ) **Linear** search

...

4. For each element *e* of *S*.[[SetData]], do

a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

TC 39

**Set.prototype. delete** ( *value* )

...

4. For each element *e* of *S*.[[SetData]], do

a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

**Set.prototype. has** ( *value* )

...

4. For each element *e* of *S*.[[SetData]], do

a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

...

core-js

**Set.prototype.add**

**Set.prototype.delete**

**Set.prototype.has**

# Solution 2 - **Rewrite Rules**

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not EMPTY and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
```

**Set.prototype.add** ( *value* )

TC 39

...

4. For each element *e* of *S*.[[SetData]], do
   a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then
   ...

# Solution 2 - **Rewrite Rules**

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not EMPTY and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
  - isSetData(base)
  - { isSetData: " _.[[SetData]] " }
```

**Set.prototype.add** ( *value* )

...

4. For each element *e* of S.[[SetData]], do
   a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then
   ...

TC 39

# Solution 2 - **Rewrite Rules**

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
  - isSetData(base)
  - { isSetData: " _.[[SetData]] " }
replace: |
  1. If SET_DATA_HAS({{ base }}, {{ value }}) is *true*, then
    1. {{ body }}
```

**Set.prototype.add** ( *value* )

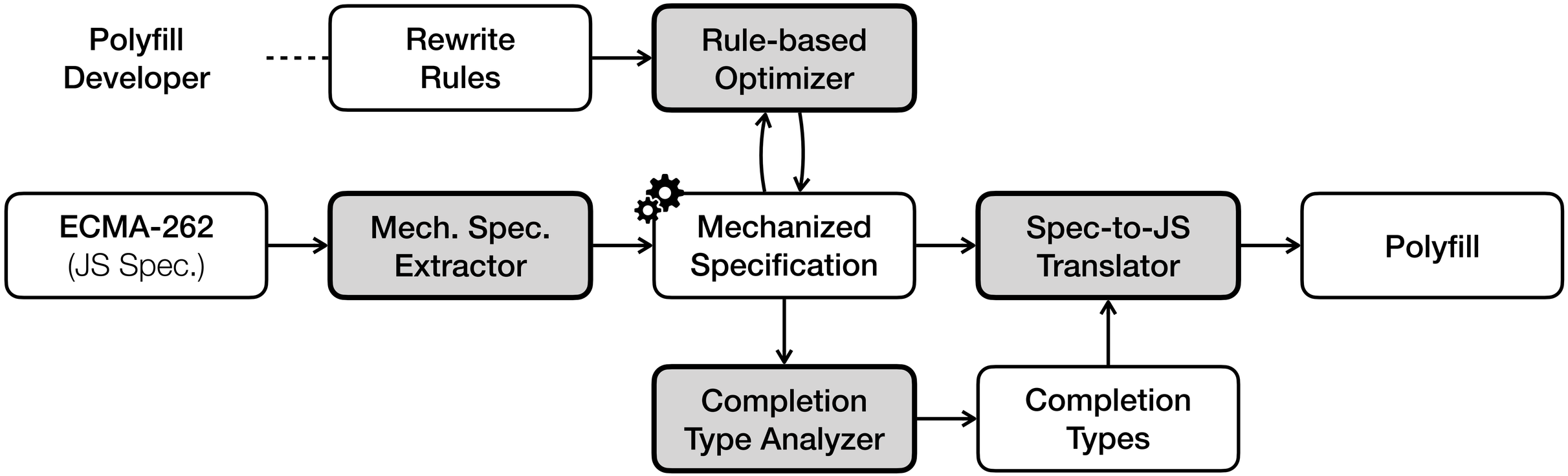
...

4. For each element *e* of *S*.[[SetData]], do
   a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then
   ...

TC 39

```
function Set.prototype.add(value) {
  ...
  if (SET_DATA_HAS(S.SetData, value)
      === true) {
    ...
  }
}
```

# **MiniPoly** - Overall Structure



# Evaluation

- Target **89 built-in APIs** in **ES2025**
- **Baselines** - `core-js` and `es-shims` (final release versions of 2025)
- Research Questions

  **RQ1) Correctness**

  **RQ2) Reduced Manual Effort**

  **RQ3) Speed**

  **RQ4) Forward Compatibility**

  RQ5) Real-World Usability

# RQ1) Correctness

**Inherently impossible** test cases passed by **Polyfill** implementation

- Tested by **Test262** - official conformance tests
- **(# of Fail) core-js - 172 / es-shims - 1,058 / MiniPoly - 16**

| | | | core-js | | es-shims† | | MiniPoly | | Minipoly-opt | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Category** | **# Built-ins** | **Tests** | **Pass (%)** | **Time** | **Pass (%)** | **Time** | **Pass (%)** | **Time** | **Pass (%)** | **Time** |
| Array | 24 | 4,713 | 4,617 (98.0) | 54.3s | 4,365 (95.1) | 81.5s | **4,713 (100.0)** | 49.2s | **4,713 (100.0)** | 51.5s |
| String | 12 | 638 | 616 (96.6) | 11.0s | 612 (95.9) | 12.4s | **638 (100.0)** | 9.7s | **638 (100.0)** | 9.5s |
| Set | 18 | 748 | 710 (94.9) | 16.1s | 698 (93.8) | 17.3s | **746 (99.7)** | 13.8s | **746 (99.7)** | 14.4s |
| Map | 12 | 322 | 298 (92.5) | 8.8s | 258 (80.1) | 8.4s | **320 (99.4)** | 7.5s | **320 (99.4)** | 7.7s |
| WeakMap | 5 | 198 | 168 (84.8) | 3.9s | – | – | **196 (99.0)** | 3.6s | **196 (99.0)** | 3.5s |
| WeakSet | 5 | 166 | 142 (85.5) | 3.9s | – | – | **164 (98.8)** | 3.8s | **164 (98.8)** | 3.1s |
| Promise | 13 | 1,264 | 1,154 (91.3) | 21.6s | 0 (0.0) | 10.9s | **1,256 (99.4)** | 18.5s | **1,256 (99.4)** | 18.9s |
| **Total** | **89** | **8,049** | **7,705 (95.7)** | **119.5s** | **5,933 (87.4)** | **130.5s** | **8,033 (99.8)** | **106.1s** | **8,033 (99.8)** | **108.7s** |

# RQ2) Reduced Manual Effort

| | | Baseline | | MiniPoly | | MiniPoly-opt | | |
|---|---|---|---|---|---|---|---|---|
| **Object** | **#** | **core-js** | **es-shims** | **Auto** | **Man.** | **Auto** | **Man.** | **Rules** |
| Array | 24 | 2,298 | 8,978 | 1,332 | 317 | | | |
| String | 12 | 1,281 | 6,885 | 418 | 130 | | | |
| Set | 18 | 2,313 | 5,880 | 864 | 190 | 861 | 283 | 114 (11) |
| WeakSet | 5 | 1,882 | – | 387 | 123 | 379 | 214 | |
| Map | 12 | 1,986 | 3,062 | 756 | 217 | 747 | 327 | 81 (6) |
| WeakMap | 5 | 2,034 | – | 446 | 115 | 436 | 231 | |
| Promise | 13 | 2,518 | 8,031 | 1,511 | 203 | | | |
| **Total**† | **89** | **14,312** | **32,836** | **5,714** | **1,295** | **5,684** | **1,705** | **195 (17)** |

3.94%

9.04%

# RQ2) Reduced Manual Effort

| | | Baseline | | MiniPoly | | MiniPoly-opt | | |
|---|---|---|---|---|---|---|---|---|
| **Object** | **#** | **core-js** | **es-shims** | **Auto** | **Man.** | **Auto** | **Man.** | **Rules** |
| Array | 24 | 2,298 | 8,978 | 1,332 | 317 | | | |
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| Promise | 13 | 2,518 | 8,031 | 1,511 | 203 | | | |
| **Total**† | **89** | **14,312** | **32,836** | **5,714** | **1,295** | **5,684** | **1,705** | **195 (17)** |

5.79%

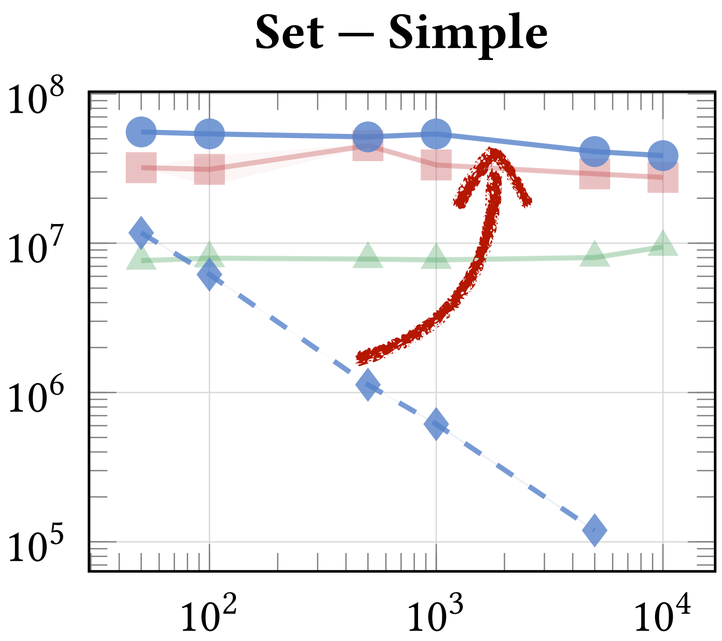
13.28%

17 rules

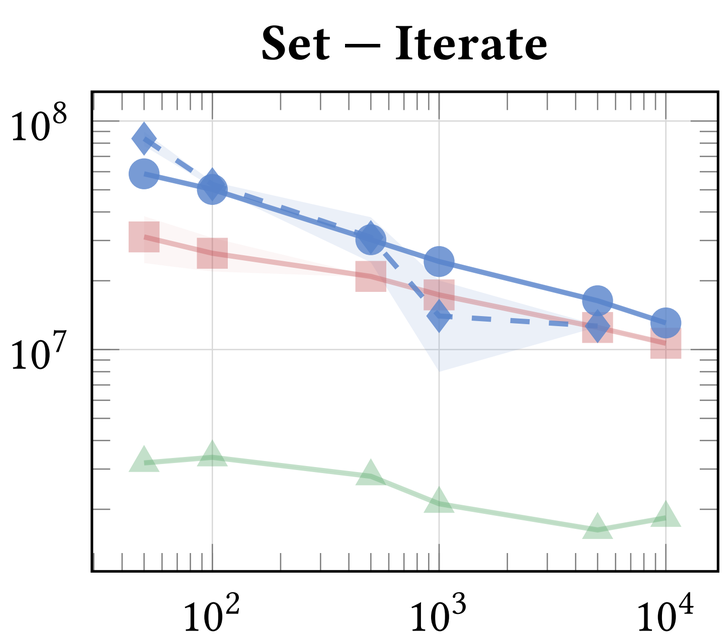
# RQ3) Speed

- We measured **throughput** (*y-axis* = ops/sec) across **sizes** (*x-axis* = N)

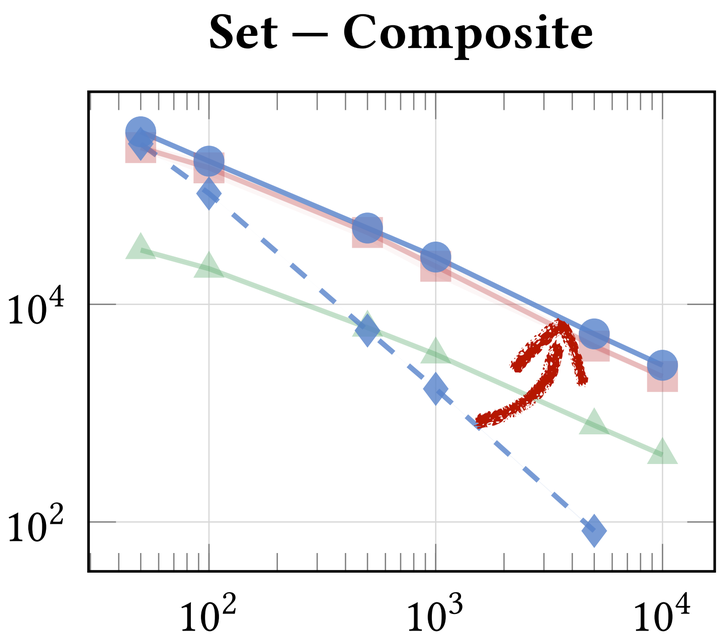
**O(N) → O(1)**




**O(N) → O(N)**



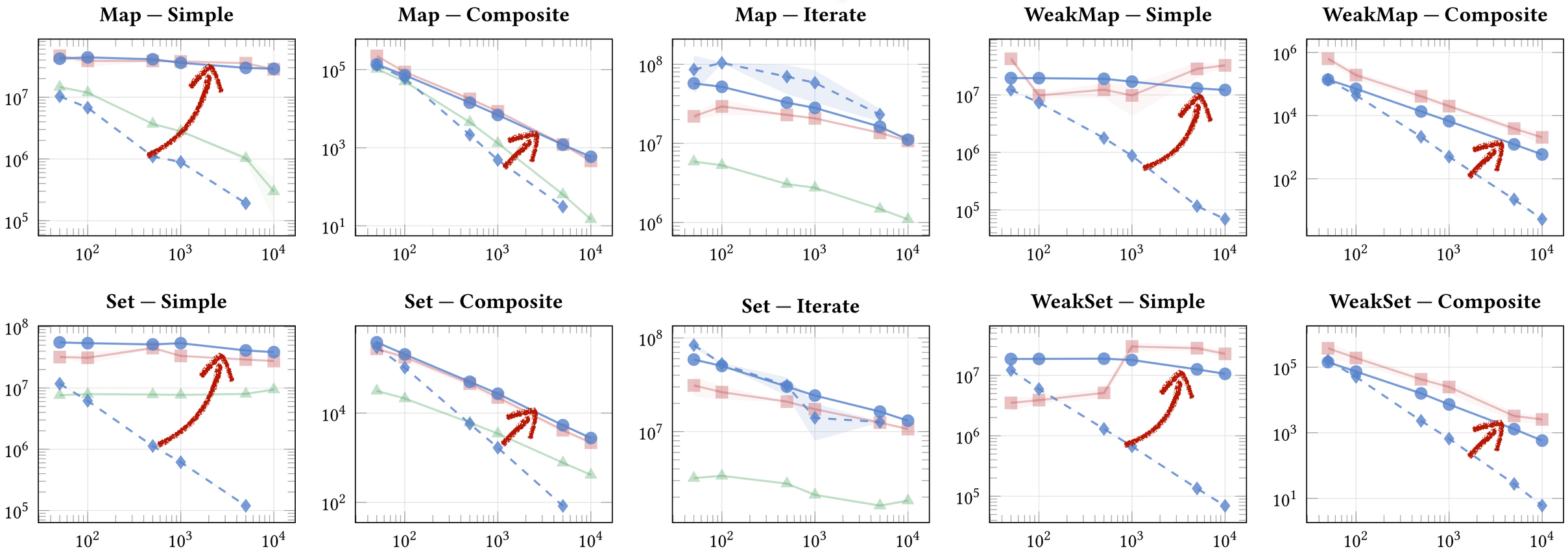

**O(N²) → O(N)**






# RQ3) Speed

**Competitive with** baselines in most cases

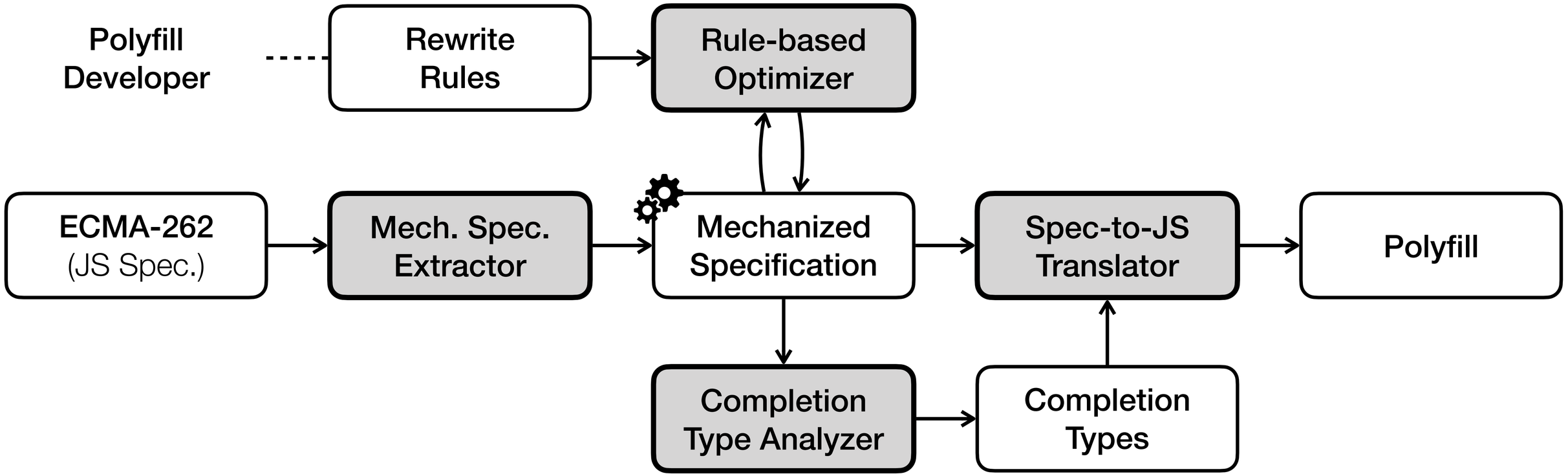


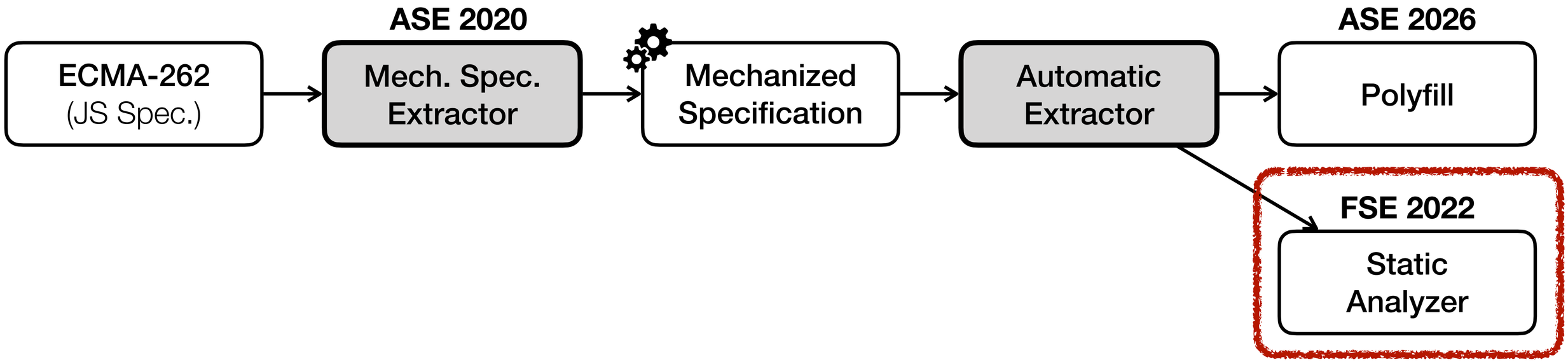
# RQ4) Forward Compatibility

- Supports Built-in APIs in **ES2026** and **Stage 2.7** with **100% correctness** and **compatible speed** with manual baselines

| | | | **Correctness (Pass %)** | | | | **Throughput (ops/sec)** — N = 5,000 | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Proposal** | **Method** | **Tests** | core-js | es-shims | MiniPoly | MiniPoly-opt | core-js | es-shims | MiniPoly | MiniPoly-opt |
| Upsert (ES2026) | Map.prototype | | | | | | | | | |
| | .getOrInsert | 28 | 26 (92.9) | 24 (85.7) | **28 (100.0)** | **28 (100.0)** | 13.7M / 2.0M | 61.2K | 107K | 8.2M / 2.5M |
| | .getOrInsertComputed | 37 | 35 (94.6) | 33 (89.2) | **37 (100.0)** | **37 (100.0)** | 11.7M / 1.8M | 40.8K | 77.7K | 6.5M / 1.9M |
| | WeakMap.prototype | | | | | | | | | |
| | .getOrInsert | 34 | 26 (76.5) | — | **34 (100.0)** | **34 (100.0)** | 18.4M / 3.1M | — | 104K | 8.2M / 2.3M |
| | .getOrInsertComputed | 43 | 32 (74.4) | — | **43 (100.0)** | **43 (100.0)** | 5.7M / 1.2M | — | 73.1K | 5.0M / 1.8M |
| Await Dictionary (Stage 2.7) | Promise | | | | | | | | | |
| | .allKeyed | 12 | — | — | **12 (100.0)** | **12 (100.0)** | — | — | — | — |
| | .allSettledKeyed | 12 | — | — | **12 (100.0)** | **12 (100.0)** | — | — | — | — |
| **Total** | | **166** | **119 (83.8)** | **57 (87.7)** | **166 (100.0)** | **166 (100.0)** | 11.3M / 1.9M | 50.0K | 89.3K | 6.8M / 2.1M |

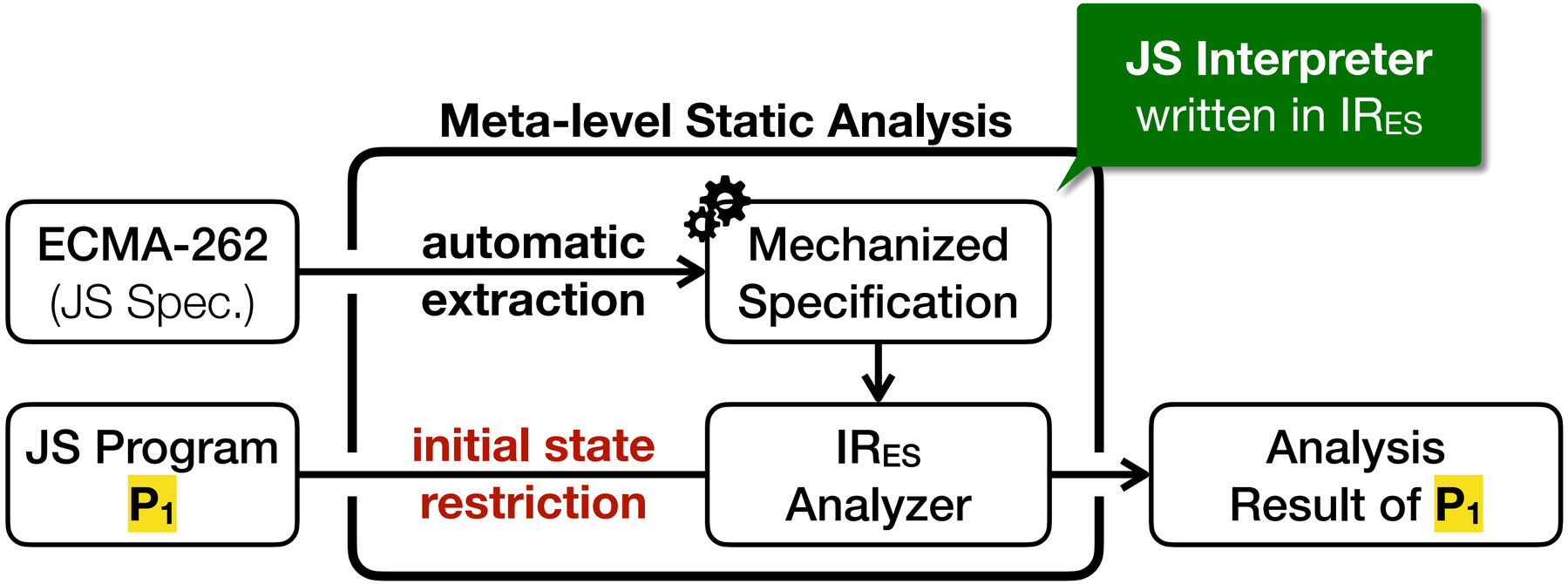
-23 -8


Polyfill
Developer
Rewrite
Rules
Rule-based
Optimizer
ECMA-262
(JS Spec.)
Mech. Spec.
Extractor
Mechanized
Specification
Spec-to-JS
Translator
Polyfill
Completion
Type Analyzer
Completion
Types


ASE 2020
ASE 2026
ECMA-262
(JS Spec.)
Mech. Spec.
Extractor
Mechanized
Specification
Automatic
Extractor
Polyfill
FSE 2022
Static
Analyzer

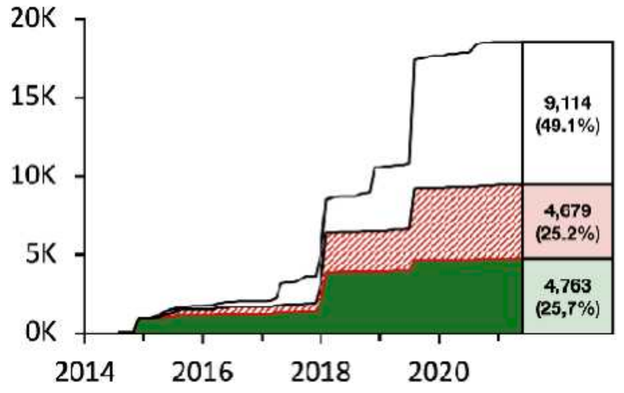
# Meta-Level Static Analysis

How to perform **static analysis** on **JavaScript programs** using **language specification**?

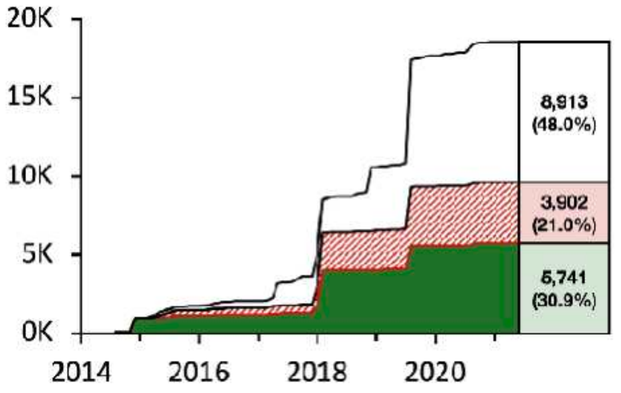


# Evaluation

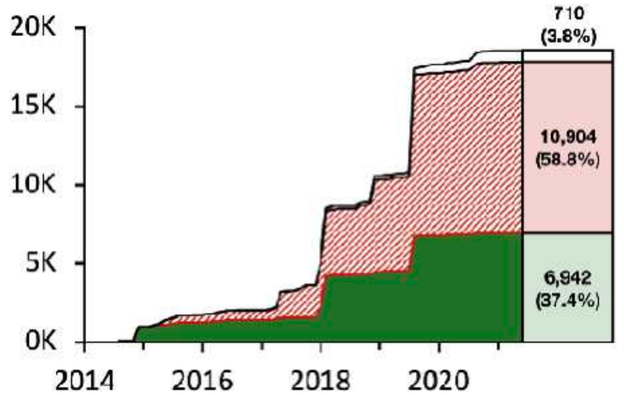
## Manual



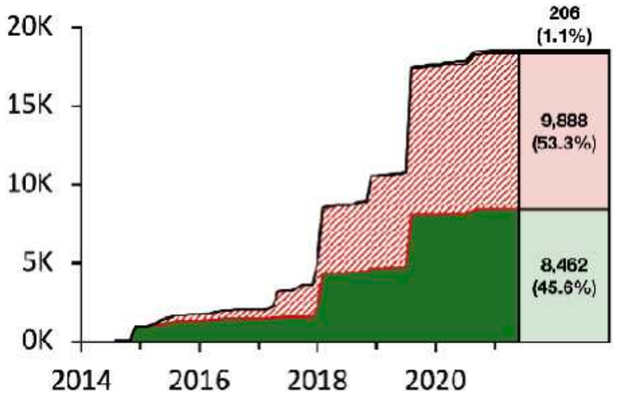

TAJS




SAFE

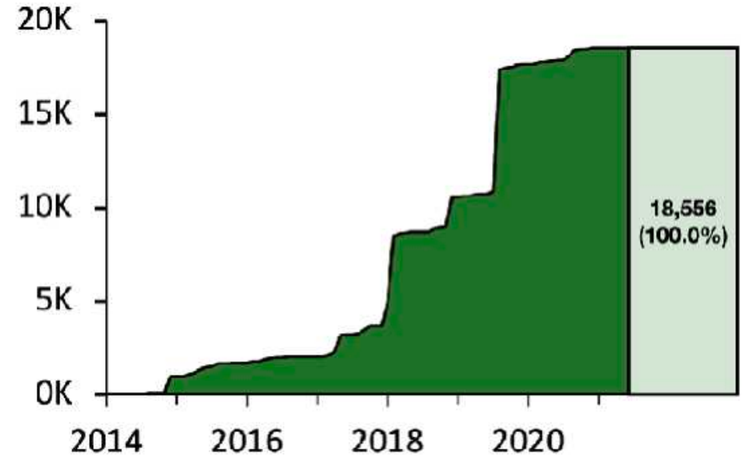



TAJS with Babel

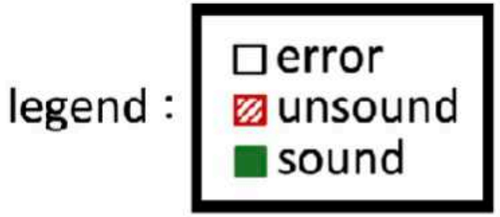



SAFE with Babel

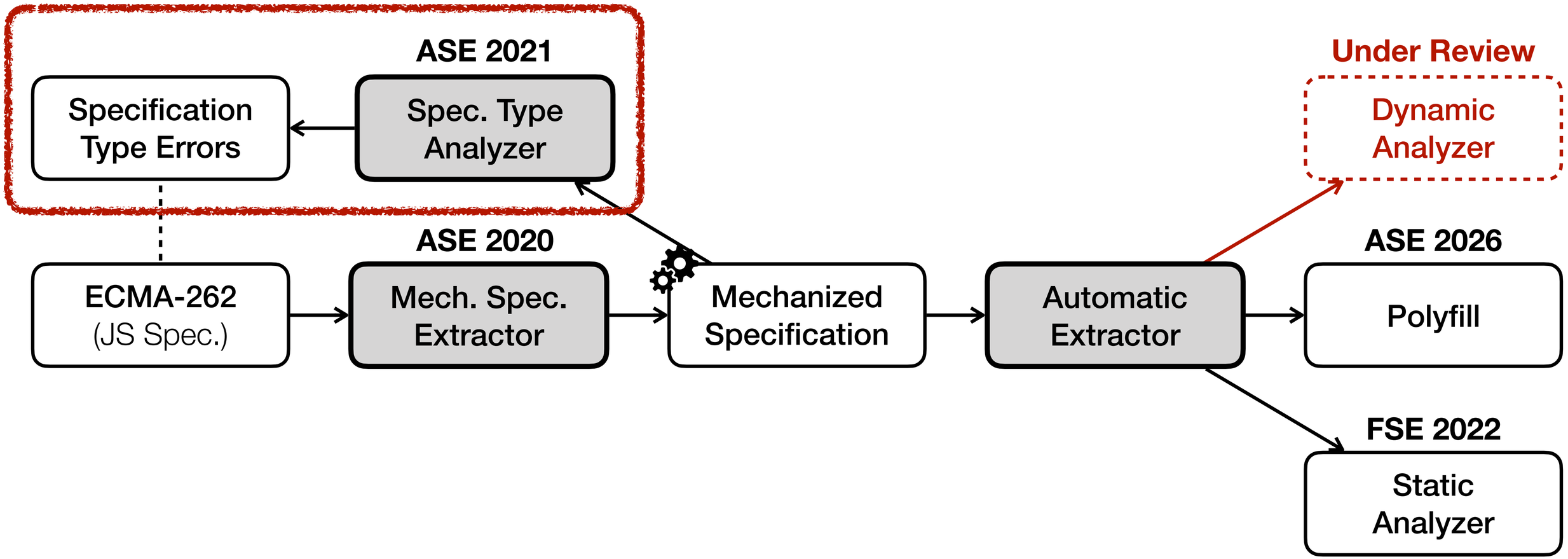
## Derived



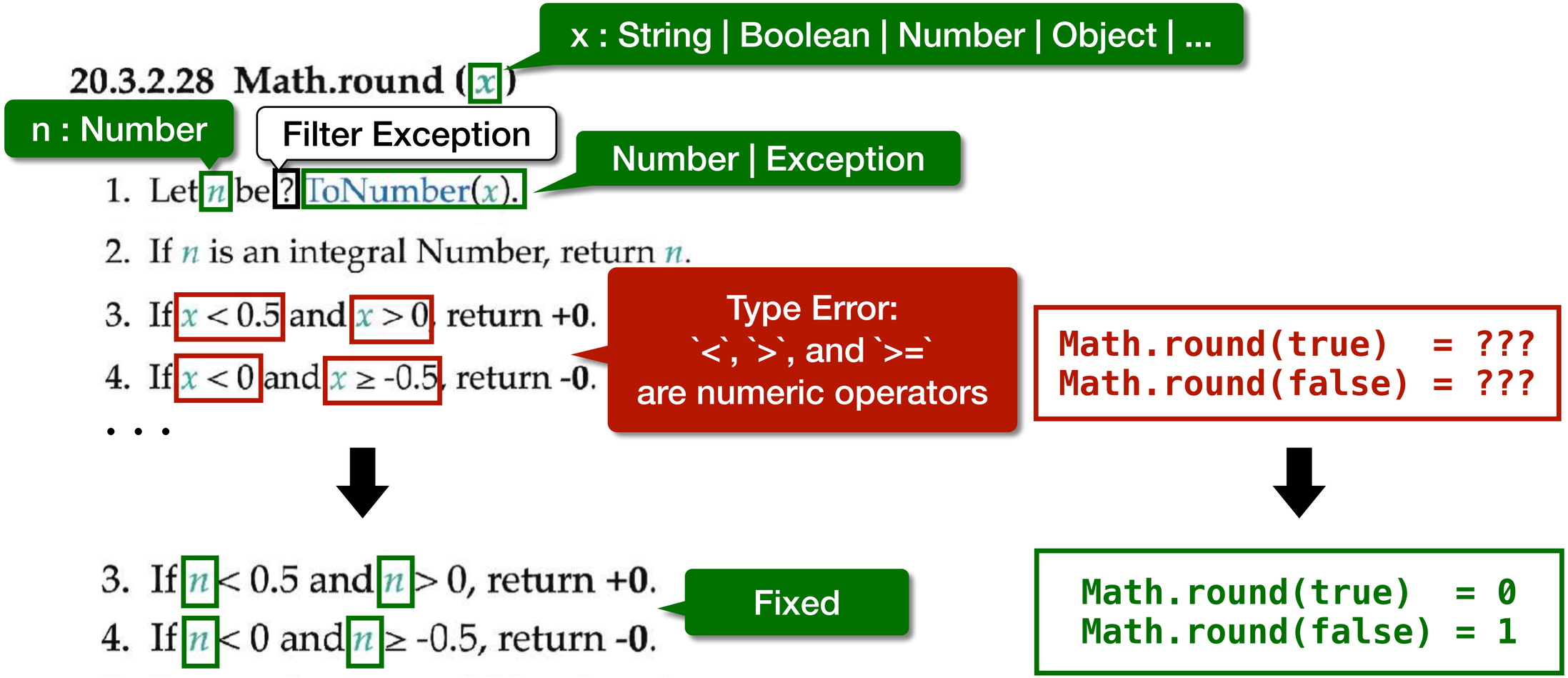

**Derived Analyzer**




x-axis : creation time (year)
y-axis : # tests


ASE 2021
Specification Type Errors
Spec. Type Analyzer
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ASE 2026
ECMA-262 (JS Spec.)
Mech. Spec. Extractor
Mechanized Specification
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Polyfill
FSE 2022
Static Analyzer

# Specification Type Analysis




https://github.com/tc39/ecma262/tree/575149cfd77aebcf3a129e165bd89e14caafc31c

# Evaluation

- Type analysis on **864 versions** of ECMA-262 in **3 years**

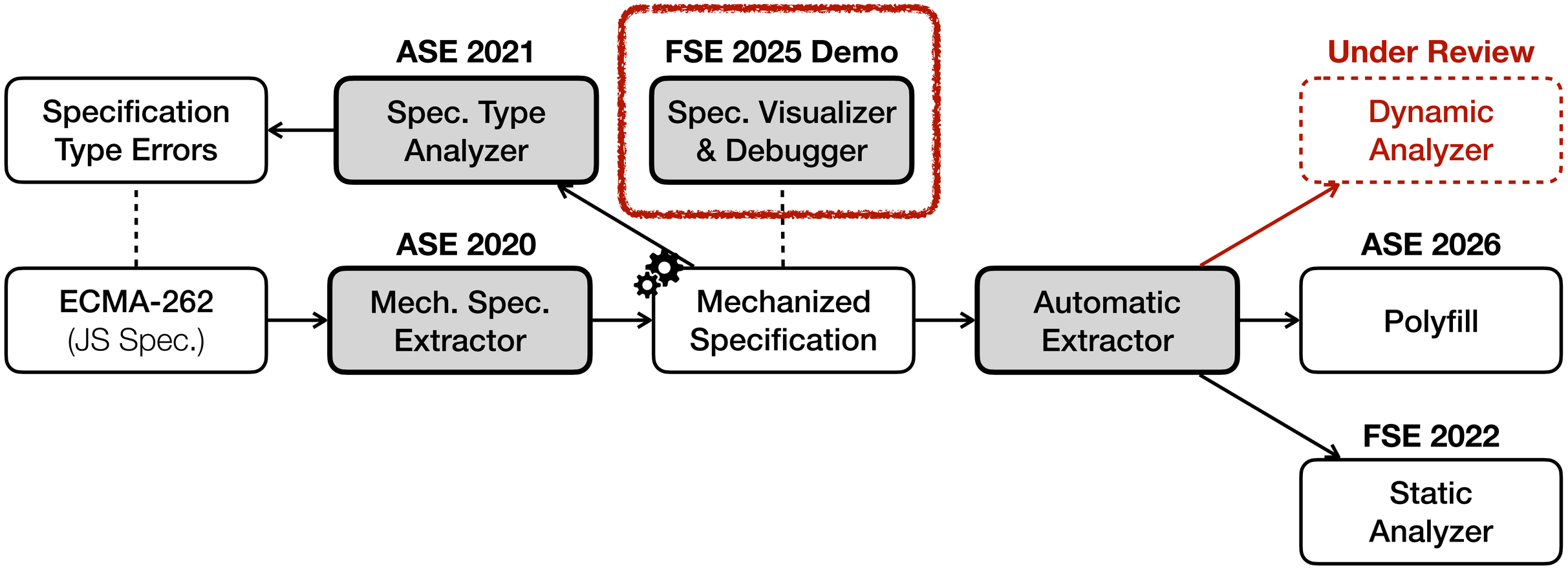
| Checker | Bug Kind | Precision = (# True Bugs) / (# Detected Bugs) | | | | | |
|---|---|---|---|---|---|---|---|
| | | no-refine | | refine | | Δ | |
| Reference | UnknownVar | 62 / 106 | 17 / 60 | 63 / 78 | 17 / 31 | +1 / -28 | / -29 |
| | DuplicatedVar | | 45 / 46 | | 46 / 47 | | +1 / +1 |
| Arity | MissingParam | | | 4 / 4 | 4 / | | / |
| Assertion | Assertion | | | 4 / 31 | 4 / 3 | | / -25 |
| Operand | NoNumber | | | 22 / 44 | 2 / | | / -59 |
| | Abrupt | | 20 / 48 | | 20 / 38 | | / -10 |
| **Total** | | 92 / 279 (33.0%) | | 93 / 157 (59.2%) | | +1 / -122 (+26.3%) | |

93 Errors Detected

59.2% Precision

| Name | Feature | # | Checker | Created | Life Span |
|---|---|---|---|---|---|
| ES12-1 | Switch | 3 | Reference | 2015-09-22 | 1,996 days |
| ES12-2 | Try | 3 | Reference | 2015-09-22 | 1,996 days |
| ES12-3 | Arguments | 1 | Reference | 2015-09-22 | 1,996 days |
| ES12-4 | Array | 2 | Reference | 2015-09-22 | 1,996 days |
| ES12-5 | Async | 1 | Reference | 2015-09-22 | 1,996 days |
| ES12-6 | Class | 1 | Reference | 2015-09-22 | 1,996 days |
| ES12-7 | Branch | 1 | Reference | 2015-09-22 | 1,996 days |
| ES12-8 | Arguments | 2 | Operand | 2015-12-16 | 1,910 days |

14 New Bugs In ES2021


ASE 2021
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Mech. Spec. Extractor
Mechanized Specification
Automatic Extractor
Polyfill
FSE 2022
Static Analyzer

ESMeta / Double Debugger Playground

Docs GitHub READY ES2026

RUN QUIT SPEC STEP OVER OUT CONTINUE SPEC BACK BACK OVER BACK OUT REWIND JS STEP

AST STEP OVER OUT

JavaScript Editor

```
1 const a = new Set
  ([1, 2, 3]);
2 const b = new Set
  ([1, 2]);
3 a.isSupersetOf(b);
```

ECMAScript Specification

Context Not Found

State Env Heap Breaks Callstack

Disabled. Start debugger to use.

JavaScript Editor

```
1 const a = new Set
  ([1, 2, 3]);
2 const b = new Set
  ([1, 2]);
3 a.isSupersetOf(b);
```

ECMAScript Specification

**Set.prototype.isSupersetOf(this, ArgumentsList, NewTarget)**

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot( *O* , [[SetData]]).
3. Let *otherRec* be ? GetSetRecord( *other* ).
4. If SetDataSize( *O* .[[SetData]]) < *otherRec* .[[Size]], return **false**.
5. Let *keysIter* be ? GetIteratorFromMethod( *otherRec* .[[SetObject]], *otherRec* .[[Keys]]).
6. Let *next* be not-started.
7. Repeat, while *next* is not done,
   a. Set *next* to ? IteratorStepValue( *keysIter* ).
   b. If *next* is not done, then
      i. If SetDataHas( *O* .[[SetData]], *next* ) is **false**, then
         i. Perform ? IteratorClose( *keysIter* ,

State Env Heap Breaks Callstack

Breakpoints USING BREAKPOINTS

search by name

| Step | Name | Enable | Remove |
|---|---|---|---|

No breakpoints. Add Breakpoint by clicking on steps in spec viewer or by searching name

RUN | QUIT | SPEC STEP | OVER | OUT | CONTINUE | SPEC BACK | BACK OVER | BACK OUT | REWIND | JS STEP

AST STEP | OVER | OUT

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         i. Perform ? IteratorClose( *keysIter* ,

State | Env | Heap | Breaks | Callstack

Specification Environment

- **ArgumentsList** : List
- **NewTarget** : undefined
- **0** : Record[Object]
- **other** : Record[Object]
- **this** : Record[Object]

JavaScript Environment

No environment variables.

RUN | QUIT | SPEC STEP | OVER | OUT | CONTINUE | SPEC BACK | BACK OVER | BACK OUT | REWIND | JS STEP

AST STEP | OVER | OUT

JavaScript Editor

```
1 const a = new Set
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  ([1, 2]);
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State | Env | Heap | Breaks | Callstack

Specification Environment

- **ArgumentsList** : List
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- **0** : Record[Object]
- **other** : Record[Object]
- **this** : Record[Object]

JavaScript Environment

No environment variables.

RUN | QUIT | SPEC STEP | OVER | OUT | CONTINUE | SPEC BACK | BACK OVER | BACK OUT | REWIND | JS STEP

AST STEP | OVER | OUT

JavaScript Editor

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State | Env | Heap | Breaks | Callstack

Specification Environment

- **ArgumentsList** : List
- **NewTarget** : undefined
- **O** : Record[Object]
- **other** : Record[Object]
- **otherRec** : Record[SetRecord]
- **this** : Record[Object]

JavaScript Environment

No environment variables.

JavaScript Editor

```
1 const a = new Set
  ([1, 2, 3]);
2 const b = new Set
  ([1, 2]);
3 a.isSupersetOf(b);
```

ECMAScript Specification

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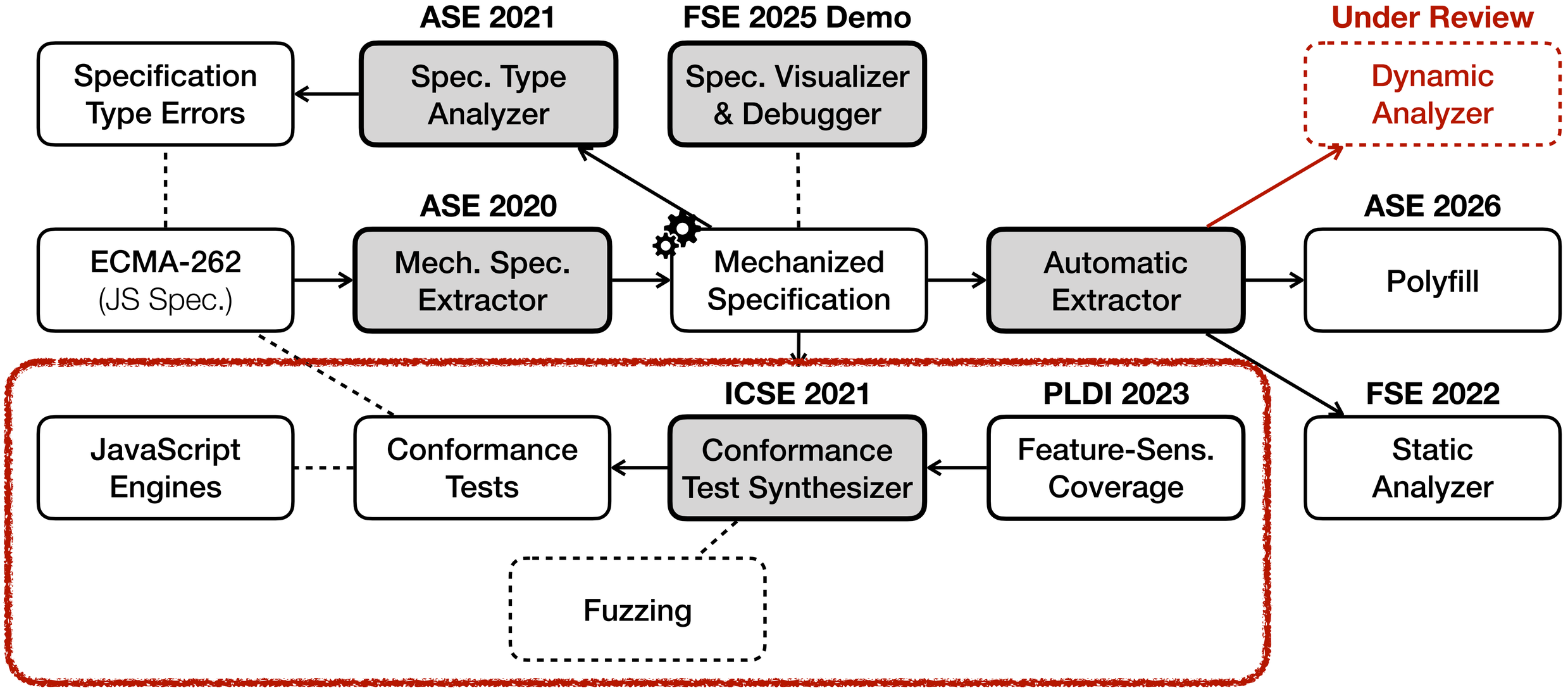
State Env Heap Breaks Callstack

Specification Environment

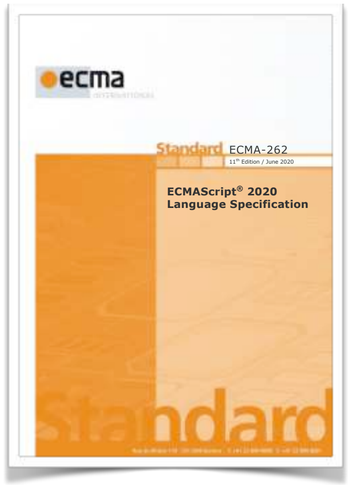
- **ArgumentsList** : List
- **NewTarget** : undefined
- **O** : Record[Object]
- **other** : Record[Object]
- **otherRec** : Record[SetRecord]
  - **Size** : 2
  - **Keys** : Record[BuiltinFunctionObject]
  - **Has** : Record[BuiltinFunctionObject]
  - **SetObject** : Record[Object]
- **this** : Record[Object]

JavaScript Environment

No environment variables.

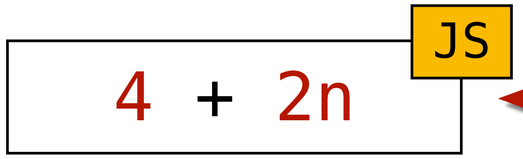

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ICSE 2021
PLDI 2023
FSE 2022
JavaScript Engines
Conformance Tests
Conformance Test Synthesizer
Feature-Sens. Coverage
Static Analyzer
Fuzzing

# **Conformance** of JavaScript Engines



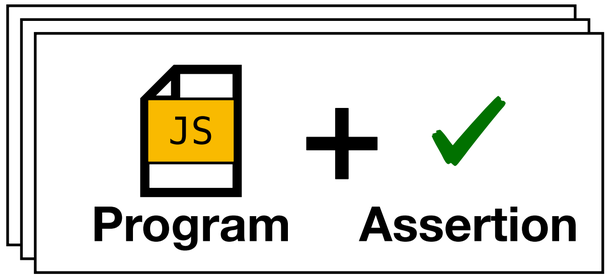

**ECMA-262**
(JavaScript Spec.)

Example:




Assertion

`TypeError`




**Conformance Tests**

**Conformance**

**Test262**
(Official Test Suite)



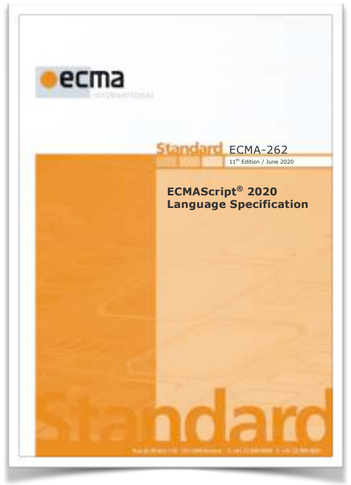
GraalVM

`QuickJS`

moddable

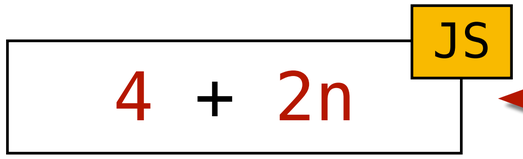
**JavaScript Engines**

# Problem - **Manual Approach**



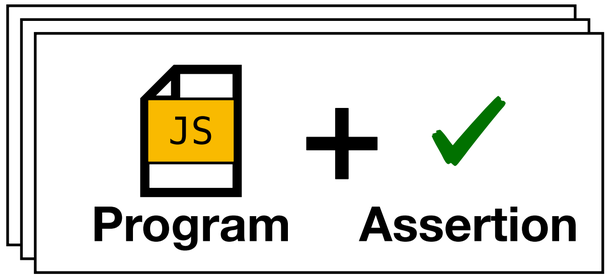

**ECMA-262**
(JavaScript Spec.)

Example:




Assertion

`TypeError`




**Conformance Tests**

**Conformance**

**Test262**
(Official Test Suite)

`Manual`



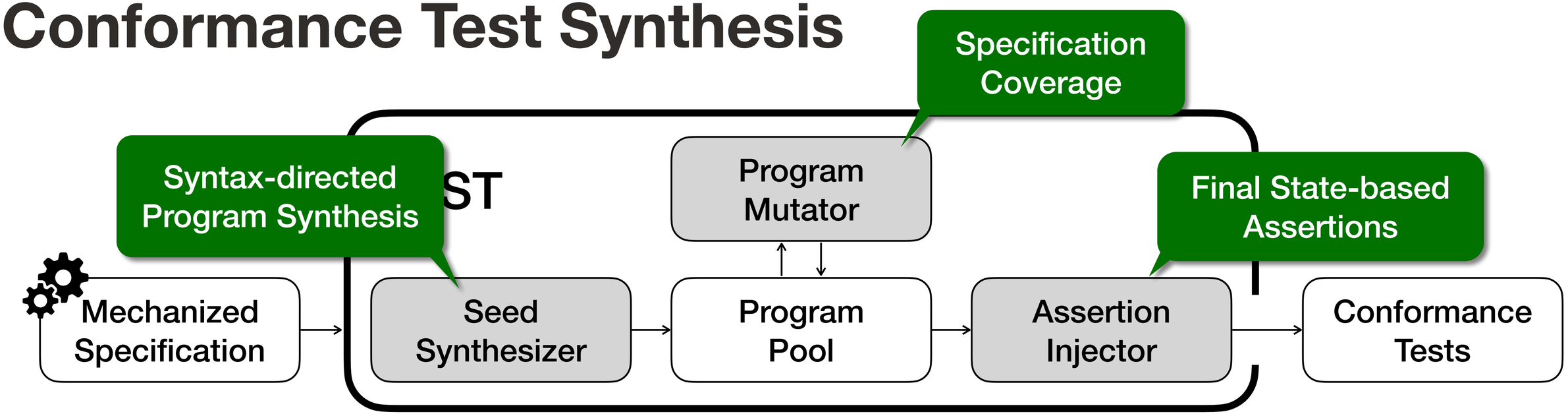
GraalVM

`QuickJS`

moddable

**JavaScript Engines**

# Conformance Test Synthesis



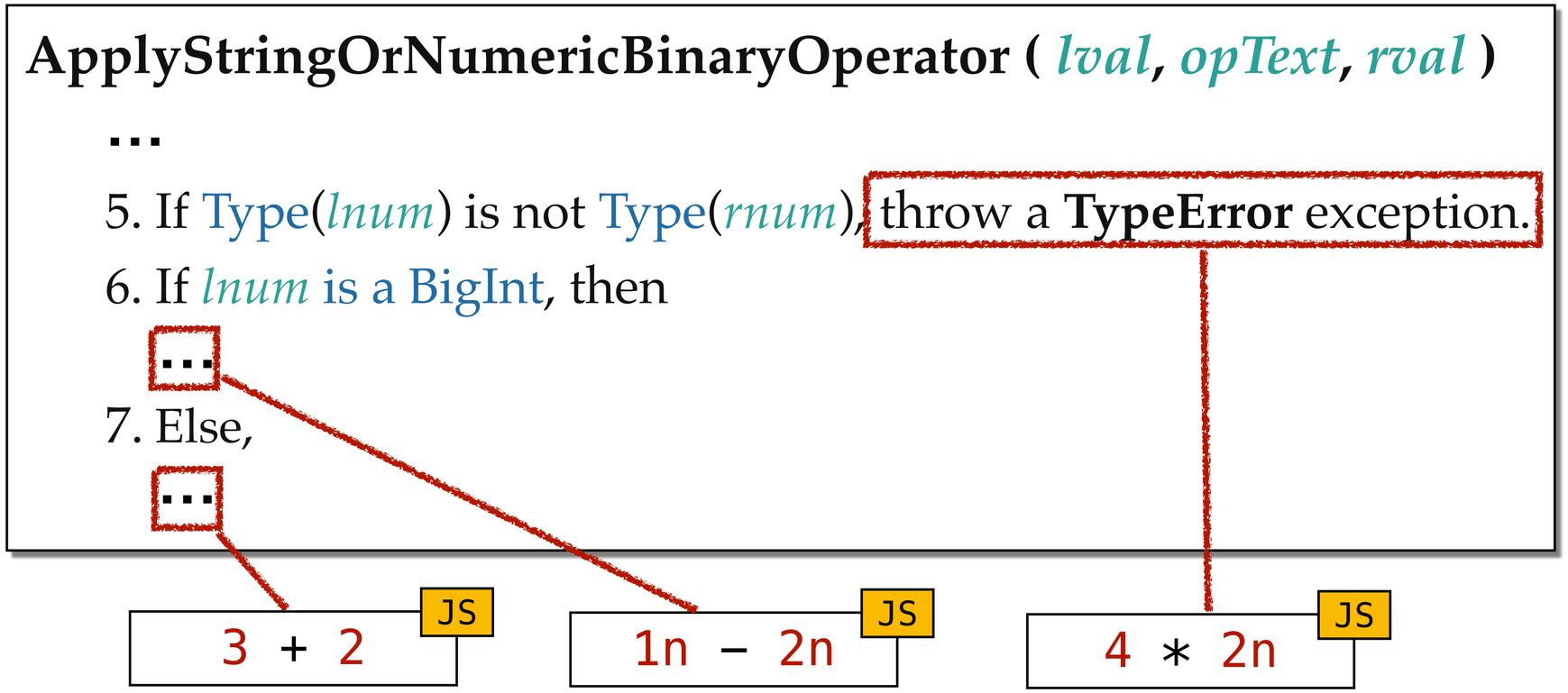

## Program Pool

```
let x = 1 + 2;
assert(x == 3);
```

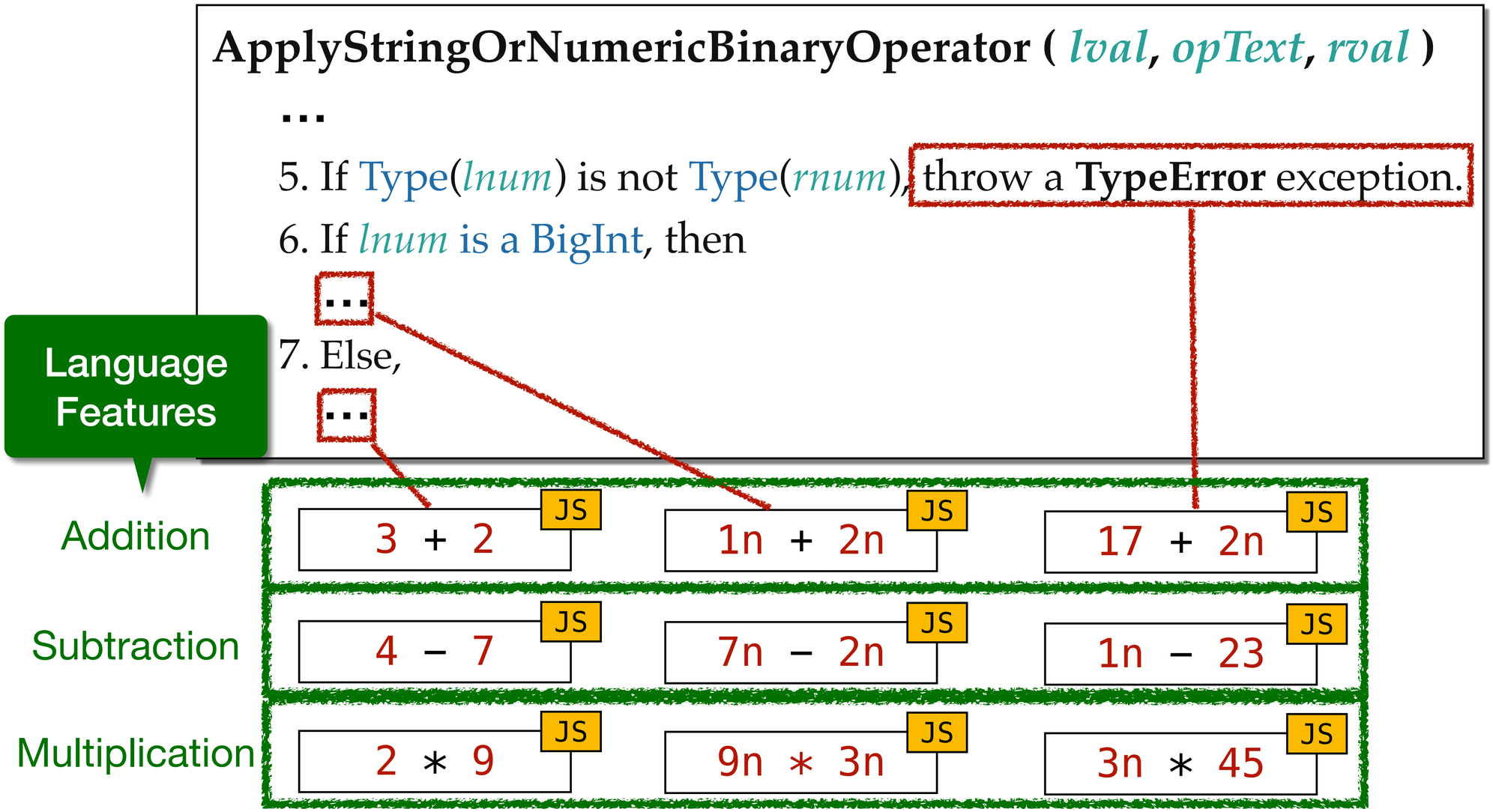
```
let x = 42;
assert(x == 42);
```

```
let x = ![];
assert(x == false);
```

# Specification Coverage



# Feature-Sensitive Coverage

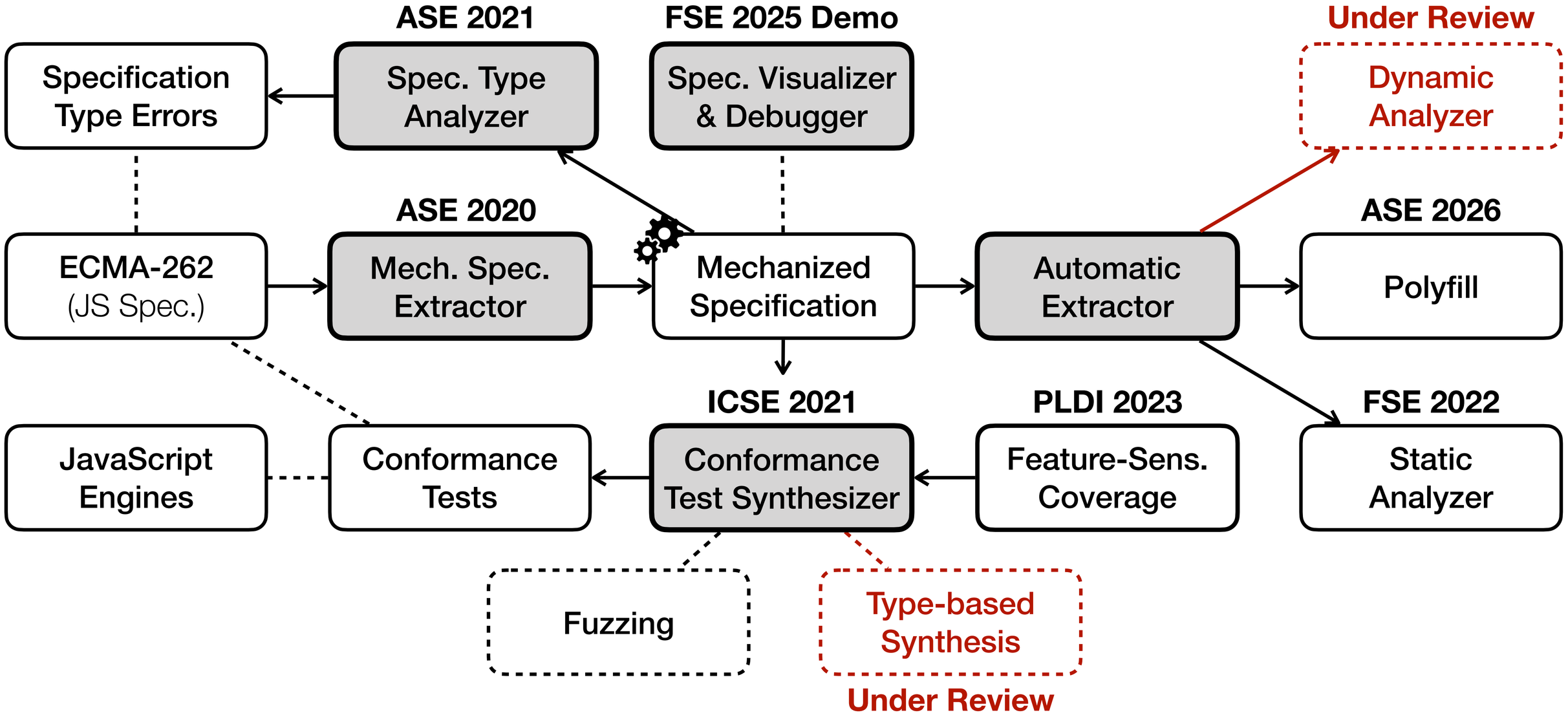


# Specification Coverage

| Kind | Name | Version | Release | # Detected Unique Bugs | | |
|---|---|---|---|---|---|---|
| | | | | # New | # Confirmed | # Reported |
| Engine | V8 | v10.8.121 | 2022.10.06 | 3 | 3 | 4 |
| | JSC | v615.1.10 | 2022.10.26 | 26 | 26 | 26 |
| | GraalJS | v22.2.0 | 2022.07.26 | 11 | 11 | 11 |
| | SpiderMonkey | v107.0b4 | 2022.10.24 | 2 | 4 | 4 |
| | **Total** | | | **42** | **44** | **45** |
| Transpiler | Babel | v7.19.1 | 2022.09.15 | 37 | 37 | 39 |
| | SWC | v1.3.10 | 2022.10.21 | 37 | 37 | 47 |
| | Terser | v5.15.1 | 2022.10.05 | 19 | 19 | 19 |
| | Obfuscator | v4.0.0 | 2022.02.15 | 1 | 2 | 7 |
| | **Total** | | | **94** | **95** | **112** |
| **Total** | | | | **136** | **139** | **157** |

42 Bugs
In Engines

94 Bugs
In Transpilers


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FSE 2025 Demo
Under Review
Specification Type Errors
Spec. Type Analyzer
Spec. Visualizer & Debugger
Dynamic Analyzer
ASE 2020
ASE 2026
ECMA-262 (JS Spec.)
Mech. Spec. Extractor
Mechanized Specification
Automatic Extractor
Polyfill
ICSE 2021
PLDI 2023
FSE 2022
JavaScript Engines
Conformance Tests
Conformance Test Synthesizer
Feature-Sens. Coverage
Static Analyzer
Fuzzing
Type-based Synthesis
Under Review

**Official tool used in CI system of ECMA-262 and Test262**






https://github.com/es-meta/esmeta

